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Key Points:

- A 3-D secondary computerized ionospheric tomography algorithm by integrating Phased Array L-band Synthetic Aperture Radar and Global Positioning System (GPS) data is proposed for the first time
- Experimental verification shows that the Faraday rotation angle can compensate the lack of GPS observation rays and improve the accuracy of the reconstructed

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Improved Computerized Ionospheric Tomography Based on GPS and PALSAR Data

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Abstract Due to factors such as the uneven distribution of ground receiving stations and the lack of effective observation rays, Global Positioning System (GPS)-based computerized ionospheric tomography (CIT) is typically of low quality and requires additional data sources. Recently, Faraday rotation angle (FRA) retrieval using the Phased Array L-band Synthetic Aperture Radar (PALSAR) full-pol data have emerged as a reliable technique for ionospheric detection. Similar to the total electron content (TEC), the FRA is the integral effect of the electron density and geomagnetic field, with the geomagnetic field being accurately estimated by the International Geomagnetic Reference Field (IGRF) model. Therefore, this paper proposes a 3-D secondary CIT algorithm by integrating PALSAR and GPS data: first, the product of the electron density values obtained from GPS-based CIT and the magnitude of geomagnetic field in corresponding voxel obtained from IGRF is used as the initial value. Then, the iterative algorithm is improved by using the FRA obtained from PALSAR data, rather than TEC, as the input for the second iteration, avoiding the approximation error caused by converting FRA into TEC. The geomagnetic field information is then separated by using the IGRF model, and the reconstructed spatial distribution is finally obtained. Experimental verification shows that the FRA can compensate for the lack of GPS observation rays to a certain extent and improve the accuracy of the reconstructed electron density. The results also indicate that the PALSAR can provide an effective and feasible data source for CIT.

Plain Language Summary The paper proposes a 3-D secondary CIT algorithm by integrating Phased Array L-band Synthetic Aperture Radar (PALSAR) and Global Positioning System (GPS) data: first, the product of the electron density values obtained from GPS-based CIT and the magnitude of geomagnetic field in corresponding voxel obtained from IGRF is used as the initial value. Then, the iterative algorithm is improved by using the FRA obtained from PALSAR data, rather than TEC, as the input for the second iteration. This avoids the approximation error caused by converting FRA into TEC. The geomagnetic field information is then separated by using the IGRF model, and the reconstructed spatial distribution is finally obtained. Experimental verification shows that the FRA can compensate for the lack of GPS observation rays to a certain extent and improve the accuracy of the reconstructed electron density. This indicates that PALSAR data can provide an effective and feasible data source for CIT.

1. Introduction

Similar to medical X-ray computed tomography (CT) (Austen et al., 1988), the computerized ionospheric tomography (CIT) utilizes radio waves transmitted by satellites, such as low-earth orbit satellite beacons and medium-to-high orbit global navigation satellite system (GNSS) data, to scan the ionosphere from the outside (Bernhardt & Siefring, 2006; Rogers et al., 2001). CIT acquires a large quantity of integrated data (i.e., total electron content (TEC)) of unknown parameters (i.e., ionospheric electron density (IED)) along intersecting paths within the scanning region. By employing appropriate reconstruction algorithms, the spatial distribution of the parameters of interest can be determined. As one of the most important parameters of the ionosphere, the spatial distribution of electron density (3-D structure of the ionosphere) is more accurate for monitoring ionospheric characteristics and quantitatively assessing the ionosphere for satellite navigation, communication, and radar applications than TEC value (2-D structure).

However, unlike medical tomography imaging, the CIT is directly affected by an uneven distribution of ground receiving stations and the lack of effective observation rays. These challenges cannot be solved solely by algorithmic improvements. CIT problems are therefore considered from the perspective of multi-data fusion in an effort to find alternative solutions (Bust & Mitchell, 2008). The reason for selecting multi-data fusion is that new data sources can provide an enormous amount of ionospheric data in different directions, thereby alleviating the ill-posed problem of CIT. Currently, third-party data sources are mainly categorized into ground-based and space-based types. First, ground-based data sources primarily come from ionosonde, which is first proposed by Garcia-Fernández et al. (2003). Dear and Mitchell (2007) optimized imaged peak electron density of multi-instrument data analysis system (MIDS) using ionosonde in mid-latitude region. Zhao et al. (2010) proposed a method that combines three-frequency beacon data with vertical and oblique measurements from the ground-based high-resolution ionospheric observation network (GHRION) in northern China, which improved the vertical resolution of tomographic imaging. However, if the tomography imaging accuracy is to be significantly improved, a dense vertical ionosonde array is required, which is costly. Measurement of occultation is the primary space-based data source (Haji et al., 1994; Rius et al., 1997). Occultation observation can improve the vertical resolution of CIT by utilizing horizontal ray information. Utilizing the Abel transform, occultation measurements are converted into IED vertical profile. However, the Abel transform assumes that the IED distribution is spherically symmetric, whereas in actuality there are horizontal gradients in the IED distribution that may lead to errors (Ma et al., 2005).

The study of Faraday rotation angle (FRA) retrieval utilizing Phased Array L-band Synthetic Aperture Radar (PALSAR) full polarimetric (full-pol) data has made significant progress in recent years (Freeman & Saatchi, 2004; Meyer & Nicoll, 2008; Rogers & Quegan, 2014; Xu et al., 2004). In ionospheric monitoring, the 2-D structure of ionosphere (i.e., FRA or TEC) obtained by the PALSAR is superior to the Global Positioning System (GPS) in both resolution and accuracy, and performs a unique role (Ji et al., 2022; Kim & Papathanassiou, 2021; Pi et al., 2011; Wang et al., 2019). Wang et al. (2023) further utilized the TEC values acquired from PALSAR to rectify the ionosonde data to enhance the authenticity of the initial values and consequently enhances the precision of GPS-based CIT. Similar to GPS, PALSAR can also retrieve the FRA at different angles, which can be used as a new data source for CIT. Therefore, this paper considers not only GPS signals but also PALSAR data (Zhao, 2024a, 2024b) to the 3-D CIT and propose a secondary tomography imaging algorithm. Here, PALSAR data is employed for the iteration process instead of improving the initial value. When the signal passes through the ionosphere twice and reaches the receiver, the full polarization PALSAR measures the change in the scattering matrix of four channels, and using an appropriate estimator, the FRA on the path can then be estimated. The FRA can be defined as the integral value of the combination of the path's IED and geomagnetic field. PALSAR directly measures FRA, whereas traditional CIT reconstructs IED using TEC as the input parameter. If FRA is converted to the TEC along the path, approximation error will occur. Fortunately, the geomagnetic field is generally stable and can be accurately obtained from the International Geomagnetic Reference Field (IGRF) model, which provides the possibility to separate electron density from FRA. In fact, Wang et al. (2017) conducted theoretical and simulation studies on 2-D CIT based on PALSAR only, but experimental validation and 3-D reconstruction are not carried out.

After determining the reconstruction area and voxel division, the initial value of the second iteration is the product value of the IED reconstructed by GPS reconstruction and the magnitude of the geomagnetic field within each voxel. Then, we process a series of PALSAR data sets within the reconstruction region to calculate the FRA values for different paths. The corresponding iterative algorithm therefore needs to be modified to obtain the distribution of the product of spatial electron density and the magnitude of the geomagnetic field. By utilizing the geomagnetic field obtained in the first step, the reconstruction result can be decomposed to obtain the IED distribution. Since PALSAR can provide a reliable and stable FRA data source, and has the characteristics of all-day, all-weather, and no ground receiving station, PALSAR can therefore be used in conjunction with GPS for tomography imaging at the same time as ground imaging, which effectively addresses the issue of insufficient rays and improves the use efficiency of SAR system. This paper will validate the proposed algorithm through both simulation and measurement data.

The outline of this paper is given as follows: In Section 2, the basics of FRA estimator based on PALSAR, and the secondary tomography imaging algorithm proposed in this paper are introduced. In Section 3, by using the full-pol PALSAR data and GPS data in Alaska, results are evaluated to show the feasibility of the proposed algorithm. Finally, our conclusions are put in Section 4.

2. Methodology

2.1. FRA Retrieval Based on Full-Pol PALSAR Data

When an electromagnetic wave travels through a magnetized plasma, such as the ionosphere, its polarization plane under-goes a rotation. The Faraday rotation angle (FRA) is a measure of the amount of such rotation that occurs and requires the knowledge of both the geomagnetic field and IED along the path. Therefore, an accurate FRA can be defined as the

$$\Omega(f) = \frac{2.365 \times 10^4}{f^2} \int_S H_0 \cos \theta_B N e ds \quad (1)$$

where $f \in [f_0 - B/2, f_0 + B/2]$ for SAR systems, B and f_0 are the bandwidth and carrier frequency, respectively. $N e$, H_0 , and θ_B are the IED, magnitude of the ambient magnetic field, and the angle between wave propagation and the magnetic field, respectively. These three parameters all exhibit variation as the path changes. In most studies (Kim et al., 2015; Meyer & Nicoll, 2008; Pi et al., 2011; Wang et al., 2019, 2023), the geomagnetic field are often approximated by its median value for convenience. When calculating the TEC, the H_0 and θ_B are outside the integral. Nonetheless, the approximation may introduce error and affect the CIT. In this paper, a stringent treatment according to Equation 1 is implemented by considering the contributions of the geomagnetic field's magnitude and θ_B during the integration procedure. When using the full-pol scattering matrix for FRA retrieval, another approximation is made, which assumes that the magnitudes of the FRA at all frequencies within the frequency range are equal to the magnitude at f_0 , that is, $\Omega(f) = \Omega(f_0)$ (Bickel & Bates, 1965; Chen & Quegan, 2010; Freeman, 2004; Li et al., 2014b; Qi & Jin, 2007), which also may lead to error, especially for future SAR system with lower carrier frequency and larger bandwidth (Li et al., 2014a). Among these FRA estimators, this paper selects the estimator based on circular polarization as having the best performance (Bickel & Bates, 1965). A more rigorous derivation of the circular polarization estimator should then be conducted in order to avoid approximations. When only the FRA is considered and after matched filter is applied in the baseband, the spectrum of signal in four channels can be defined as

$$S_{o,pq}(f_b) = S_r(f_b) \cdot Z_{pq}(f) \cdot H(f_b) \quad (2)$$

where $f_b = f - f_0$ is the frequency in the baseband. p and q can either be left- (l) or right-circular (r) polarization. $S_r(f_b)$ is the ideal echo spectrum of delay, that is,

$$S_r(f_b) = \text{rect}[f_b/B] \exp(-j\pi f_b^2/K) \exp(-j2\pi f_b t_0) \quad (3)$$

where t_0 is the delay between the radar at a certain azimuth position and the target under ideal condition, K is the linear FM rate and $H(f_b)$ is the corresponding matched filter

$$H(f_b) = \text{rect}[f_b/B] \exp(j\pi f_b^2/K) \quad (4)$$

$Z_{pq}(f)$ is the measured scattering matrix in circular basis affected by the FRA and can be written as

$$\begin{cases} Z_{ll}(f) = S_{hh} - S_{vv} + 2jS_{hv} \\ Z_{lr}(f) = (S_{hh} + S_{vv}) \exp(j\pi/2 - j2\Omega(f)) \\ Z_{rl}(f) = (S_{hh} + S_{vv}) \exp(j\pi/2 + j2\Omega(f)) \\ Z_{rr}(f) = S_{vv} - S_{hh} + 2jS_{hv} \end{cases} \quad (5)$$

where $S_{hh}, S_{vh}, S_{hv}, S_{vv}$ are the true scattering matrix of linear horizontal (h) and vertical (v) polarizations. Substituting Equations 3–5 into Equation 2 and ignoring the amplitude changes and t_0 , the compressed signal of four channels can be given by the inverse Fourier transform of $S_{o,pq}(f)$, that is,

$$\begin{cases} S_{ll} = S_{hh} - S_{vv} + 2jS_{hv} \\ S_{lr} = (S_{hh} + S_{vv})e^{j\pi/2} \int_{-B/2}^{B/2} e^{-j2\Omega(f)} e^{j2\pi f_b t} df_b \\ S_{rl} = (S_{hh} + S_{vv})e^{j\pi/2} \int_{-B/2}^{B/2} e^{j2\Omega(f)} e^{j2\pi f_b t} df_b \\ S_{rr} = S_{vv} - S_{hh} + 2jS_{hv} \end{cases} \quad (6)$$

where $2\Omega(f)$ will introduce errors (i.e., polarimetric dispersion (Li et al., 2014a)) in the cross-polarization channel, which can be expanded in a Taylor series at f_0

$$2\Omega(f) = 2\Omega(f_0) \sum_{k=0}^{+\infty} (-1)^k (k+1) (f_b/f_0)^k \quad (7)$$

Ignoring third-order and higher terms, Equation 7 is approximated as

$$2\Omega(f) \approx 2\Omega(f_0) - 4\Omega(f_0)f_b/f_0 + 6\Omega(f_0)f_b^2/f_0^2 \quad (8)$$

If only the constant phase error of Equation 8 is considered, that is, $\Omega(f) = \Omega(f_0)$, Equation 8 will degenerate into the traditional form (Bickel & Bates, 1965). According to the property of the inverse Fourier transform, the linear term of Equation 8 will cause image shift, the relative shift between S_{lr} and S_{rl} is

$$\Delta r = 2c\Omega(f_0)/\pi f_0 \quad (9)$$

According to the parameters of the PALSAR system, even if the FRA reaches 360° , the relative shift is less than 1m, which is significantly less than the range sampling spacing (Pi et al., 2011). The image phase of the cross-polarization channels can be considered as phase-preserving and have no impact on the FRA retrieval. The quadratic term of Equation 8 will cause image defocusing, which can be represented as quadratic phase error (QPE), that is,

$$QPE = (3B^2/2f_0^2) \cdot \Omega(f_0) \quad (10)$$

Similarly, QPE is significantly lower than the 45° threshold for PALSAR even when the FRA is considerable, so its defocusing effect on the cross-polarization channel can also be ignored. Nonetheless, the QPE introduces an additional constant phase error at the peak of the pulse compression, which is one-third of the QPE and may affect the FRA retrieval (Cumming & Wong, 2004). In conclusion, the linear and quadratic terms in Equation 8 can be ignored (i.e., image shift and defocusing), but the constant term of Equation 8 and the constant phase error introduced by the QPE need to be considered. Therefore, Equation 6 can be rewritten as

$$\begin{cases} S_{ll} = S_{hh} - S_{vv} + 2jS_{hv} \\ S_{lr} = (S_{hh} + S_{vv}) \exp(j\pi/2 - j2\Omega(f_0) - jQPE/3) \\ S_{rl} = (S_{hh} + S_{vv}) \exp(j\pi/2 + j2\Omega(f_0) + jQPE/3) \\ S_{rr} = S_{vv} - S_{hh} + 2jS_{hv} \end{cases} \quad (11)$$

Thus, Ω' can be obtained by the following conjugation processing

$$\Omega' = -\frac{1}{4} \arg(S_{lr} \cdot S_{rl}^*) = (4f_0^2 + B^2)/4f_0^2 \cdot \Omega(f_0) \quad (12)$$

The FRA retrieval at f_0 can then be derived by

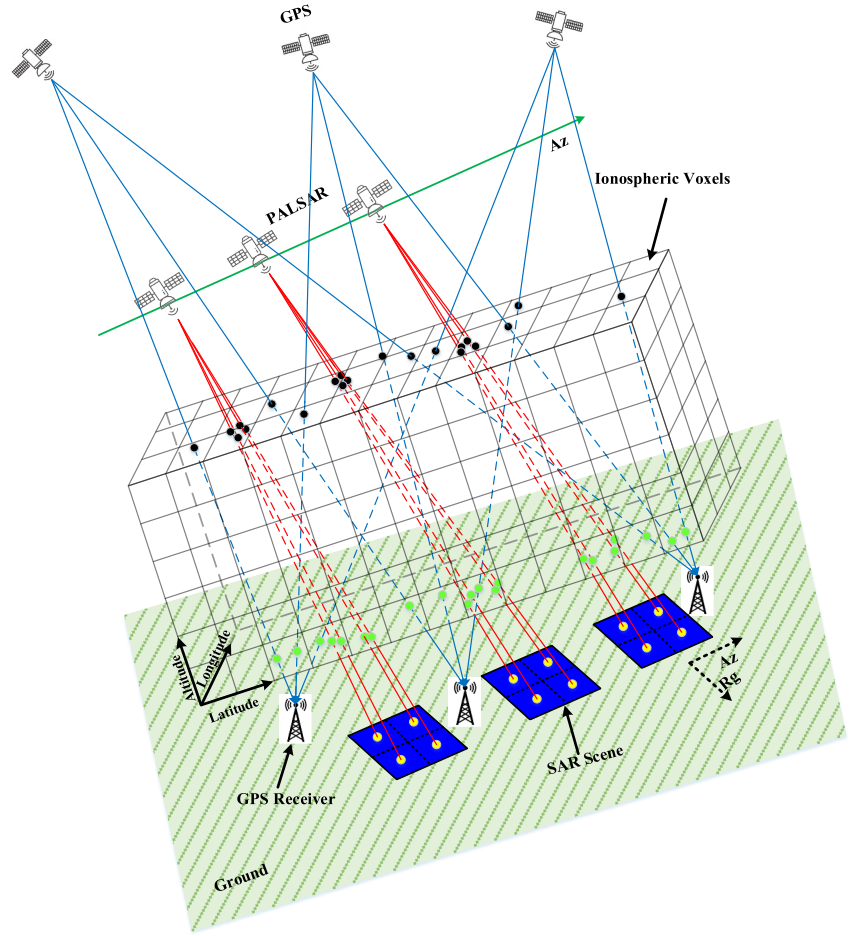


Figure 1. Ionospheric tomography diagram based on GPS and PALSAR data.

$$\Omega_r(f_0) = \frac{4f_0^2 \Omega'}{4f_0^2 + B^2} \quad (13)$$

2.2. GPS-Based CIT Theory

GPS-based CIT is essentially the reconstruction of the spatial electron density distribution based on the TEC values obtained from different angles. The slant TEC over a special GPS path l is described as

$$\text{STEC} = \int_l Ne(\mathbf{r}) ds \quad (14)$$

where $Ne(\mathbf{r})$ is the IED along the path to be reconstructed. $\mathbf{r}$ is the position vector composed of longitude, latitude, and altitude. As shown in Figure 1, the ionosphere is divided into J voxels along the longitude, latitude, and altitude.

For GPS-based CIT, each voxel contains a constant IED value. The GPS rays are represented by blue lines, and the intersection points where each ray entries and exits the ionosphere are denoted by the black and green solid dots, respectively. In addition, it is necessary to select valid rays for reconstruction, which is defined as the ray that passes through the entire reconstruction area (i.e., the entry and exit points of the ray have intersections with the upper and lower surfaces of the reconstruction area, respectively). Therefore, it is necessary to select appropriate basis functions b_j to characterize the IED, that is,

$$Ne(\mathbf{r}) \approx \sum_{j=1}^J x_j b_j(\mathbf{r}) \quad (15)$$

where x_j is the electron density in the j th voxel, and

$$b_j(\mathbf{r}) = \begin{cases} 1, & \text{if } \mathbf{r} \text{ is inside } j\text{th voxel} \\ 0, & \text{otherwise} \end{cases} \quad (16)$$

For GPS raypath's TEC measurements, substituting Equations 15 and 16 into Equation 14 yields

$$STEC_i = \sum_{j=1}^J x_j \int_l b_j(\mathbf{r}) ds = \sum_{j=1}^J a_{ij} x_j, (i = 1, 2, \dots, M) \quad (17)$$

where M represents the total number of slant TEC observations or ray paths, and a_{ij} is the intercept of the i th ray in the j th voxel. Therefore, Equation 17 can be put in a matrix form as

$$\mathbf{y}_{M \times 1} = \mathbf{A}_{M \times J} \cdot \mathbf{X}_{J \times 1} \quad (18)$$

where $\mathbf{y}$ is a column vector consisting of M TEC observation values, $\mathbf{A}$ is a $M \times J$ matrix representing the intercepts of the rays within the corresponding pixels, and $\mathbf{X}$ is a column vector consisting of J electron density voxels values.

As shown in Figure 1, the GPS-based ionospheric tomography is a rank-deficient problem due to the factors including the uneven distribution of GPS receivers, sparse GPS rays within the reconstruction area, and restricted observation angles. The multiplicative algebraic reconstruction technique (MART) algorithm is widely used in CIT to address the rank-deficient problem. The MART algorithm gives an initial value of IED, $\mathbf{X}^{(0)}$, to each voxel using an empirical model (the Empirical Canadian High Arctic Ionospheric Model (E-CHAIM) model is used in this paper) before iteration, and then gradually improves the accuracy of IED within the reconstruction area through iterative method. The fundamental iterative equation is (Bust & Mitchell, 2008; Raymund et al., 1990)

$$\mathbf{X}_j^{(k+1)} = \mathbf{X}_j^k \left(\frac{y_i}{\langle \mathbf{X}^k, a_i \rangle} \right)^{\frac{\lambda_k a_{ij}}{\|a_i\|}} \quad (19)$$

where $\mathbf{X}_j^{(k+1)}$ is the IED of the j th voxel at iteration $k + 1$, and λ ranges from 0 to 1. $\langle \cdot \rangle$ and $\| \cdot \|$ denote inner product and norm, respectively. The iteration order is to use the i th STEC measurement to perform the $(k + 1)$ th correction for each voxel in turn, and after M corrections, a round of iteration is completed, and the results are finally converged to the solution of Equation 18.

2.3. Improved CIT Algorithm Based on Full-Pol PALSAR

Ganguly et al. (2001) first demonstrate the potential application of FRA in tomography imaging based on the data from the satellites of Explorer 22. According to Equation 1 and discussion in Section 2.1, the full-pol PALSAR can also retrieve the FRA, which is the integrated value of the IED and the geomagnetic field vector along the SAR path, allowing us to reconstruct the electron density from various FRA angles. As shown in Figure 1, the PALSAR moves along the azimuth direction and images the ground, the radar images of PALSAR can then be obtained. As indicated by the blue region on the ground, a typical full-pol PALSAR scene consists of $1,248 \times 18,432$ pixels in range and azimuth dimensions. In order to obtain more ray information (i.e., FRA measurement value), we divide each scene data into four parts. On the basis of the procedure described in Section 2.1, which separately processes each sub-scene, the FRA value of the ray (red line) from the satellite to the center point of each sub-scene (solid yellow point) can be retrieved by Equation 13. According to Equation 1, Equation 13 can also be defined as

$$\Omega_r(f_0) = \frac{2.365 \times 10^4}{f_0^2} \int_l H_0 \cos \theta_B N ds \quad (20)$$

Within each voxel, both H_0 and Ne can be considered as constant, and H_0 can be accurately obtained from the IGRF model. Similarly to a_{ij} , $\cos \theta_B$ is also related to both the voxel and the ray. Therefore, under the same ionospheric voxel conditions, J voxels, the FRA measurement value of Equation 20 can be represented as

$$\Omega_{ri}'(f_0) = \frac{2.365 \times 10^4}{f_0^2} \sum_{j=1}^J a_{ij}' x_j', (i' = 1, 2, \dots, M') \quad (21)$$

There are M' FRA observations or PALSAR ray paths. $x_j' = x_j \cdot H_{0j}$ and H_{0j} is the magnitude of the geomagnetic field in voxel j . $a_{ij}' = a_{ij} \cdot \cos \theta_{B_{ij}'}$, and a_{ij}' is the intercept of the i th PALSAR ray in the j th voxel. It can be seen that $\cos \theta_{B_{ij}'}$ acts as weighting function for the a_{ij}' . The matrix form of Equation 21 can be written as

$$\mathbf{y}_{M' \times 1}' = \mathbf{A}_{M' \times J}' \cdot \mathbf{X}_{J \times 1}' \quad (22)$$

Similar to GPS-based CIT, the solution of Equation 22 can still be solved by Equation 19, but the result of the reconstruction is the spatial distribution of the product of IED and H_0 . The IED can finally be reconstructed by utilizing the previously obtained B_0 from the IGRF model. It should be noted that we use the product of the output of GPS-based CIT and H_0 as the initial value of PALSAR-based CIT. Figure 2 depicts the corresponding flowchart of the proposed CIT algorithm.

3. CIT Experiments

Due to the sparsity of rays in GPS-based tomography imaging, not each voxel can be traversed by rays, causing some voxels to remain unchanged in IED before and after CIT. On the other hand, PALSAR supplements the ray density of tomographic imaging from different angles, increasing the number of voxels that are traversed (or corrected) by rays, thereby enhancing the tomographic accuracy to some extent. Based on semi-physical simulation and measured data, this section validates a comparison of tomographic imaging accuracy before and after the addition of PALSAR data (Zhao, 2024a) and conducts preliminary analysis. The algorithm software on which this paper is based are available in Zhao (2024b).

3.1. Simulation Experiment

In order to validate the viability of the proposed secondary CIT algorithm, the latitude range of 60° to 65°N and the longitude range of 145° to 152°W are chosen. 1° and 1° are the discretized intervals for latitude and longitude, respectively. The height range is 100–600 km with a 10 km interval. Thus, the ionosphere is divided into $5 \times 7 \times 50$ voxels. Similar to Figure 1, the reconstructed area of the horizontal perspective is shown in Figure 3, which has 6 GPS receivers (solid red point), 7 positions of PALSAR satellite (blue pentagram) corresponding to 7 radar images, 28 entry (black square) and exit (green square) points of the PALSAR rays passing through the ionosphere.

Using the E-CHAIM model, the IED distribution of 07:00 and 07:28 UT on 10 March 2011 are selected as the initial value and the assumed true value, respectively. The true value, Ne_T , is shown in Figure 4, which illustrates the distribution of IED with latitude and altitude on the longitude surfaces of 145.5°W, 146.5°W, 147.5°W, 148.5°W, 149.5°W, 150.5°W, and 151.5°W. They respectively represent the IED distribution of the voxels between 145°W and 146°W, 146°W and 147°W, 147°W and 148°W, 148°W and 149°W, 149°W and 150°W, 150°W and 151°W, and 151°W and 152°W. Then, the actual coordinates of GPS and ground receivers within 5 min of the true value time (i.e., between 07:26 and 07:30 UT on 10 March 2011) are read from the GPS data (los_20110310.001.h5 and site_20110311.001.h5 files) (CEDAR, 2024). During this period, a total of 493 effective rays are obtained from the 6 GPS receivers in the reconstruction area. Based on above voxel, GPS information and the true value, the coefficient matrix $\mathbf{A}$ can be calculated, and the simulated slant true TEC values of each GPS rays are then calculated by Equation 18. The GPS-based iterative reconstruction is performed using Equation 19, and the CIT results on the 7 longitude surfaces, that is, Ne_{GPS} are shown in Figures 5a1–5a7.

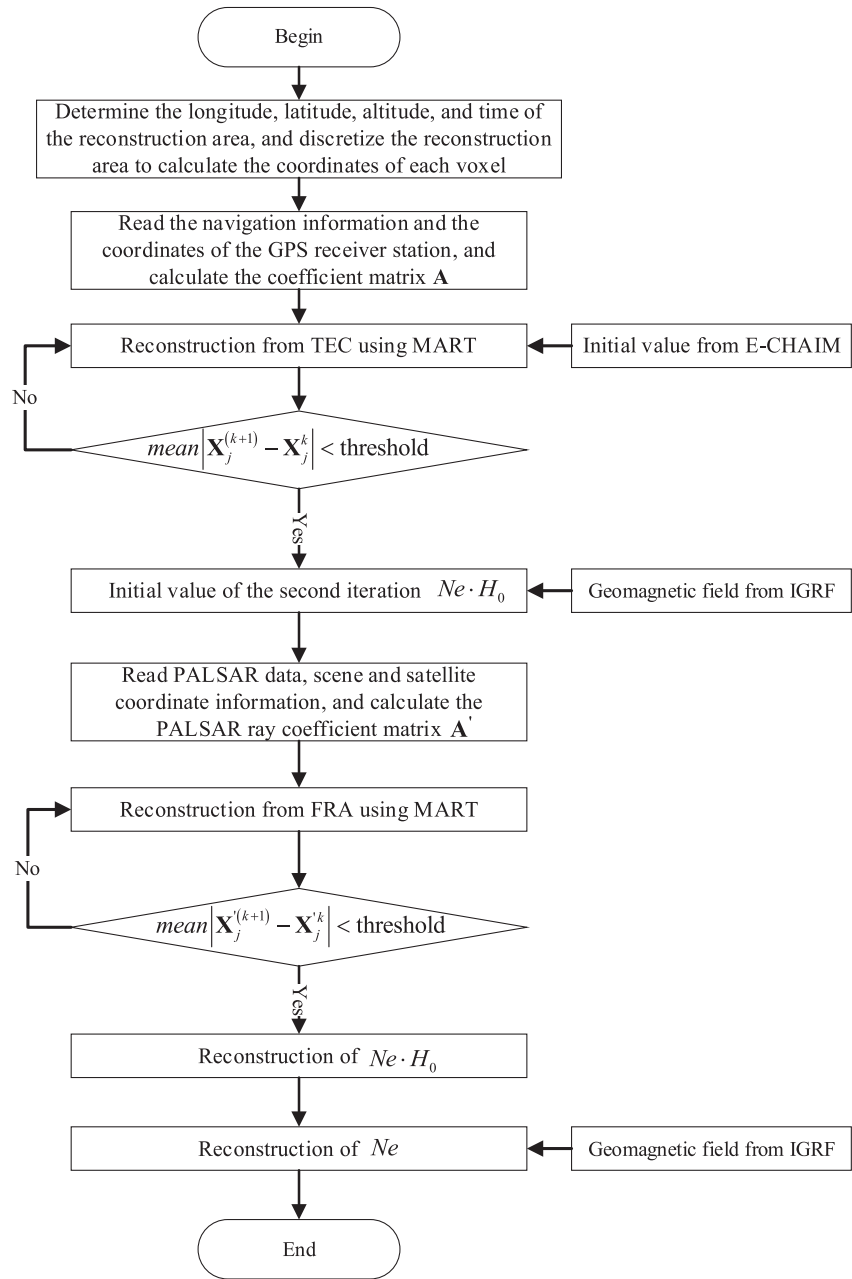


Figure 2. The flowchart of the proposed computerized ionospheric tomography algorithm.

The product of the H_0 generated from IGRF for each voxel and Ne_{GPS} , will be used as the initial value for the next iteration. As shown in Figures 3 and 7 radar images of full-pol PALSAR data (Data name: ALPSRP276021230 to ALPSRP276021290) along ascending orbit are selected for semi-physical simulation validation. Each SAR scene is divided into 4 sub-scenes, and each sub-scene contains $312 \times 4,608$ pixels in range and azimuth dimensions. Base on the information in the “ceos_leader” and “workreport” files of each PALSAR data, the coordinates of 7 PALSAR positions and the center point of the sub-scene can be obtained. Based on the above settings, the intercept of each PALSAR ray in the voxels can be calculated. Using the IGRF model's geomagnetic field distribution, the angle between the geomagnetic field vector and PALSAR ray in the traversed voxel can be calculated, and the angle multiplied by the intercept is A' . The assumed true FRA can then be calculated by A' , and $H_0 \cdot Ne_T$ according to Equation 22. In the simulation, the reciprocal is assumed by performing a symmetrization operation on each sub-scene data (Li et al., 2014b), which can be considered calibrated to the ideal case, and

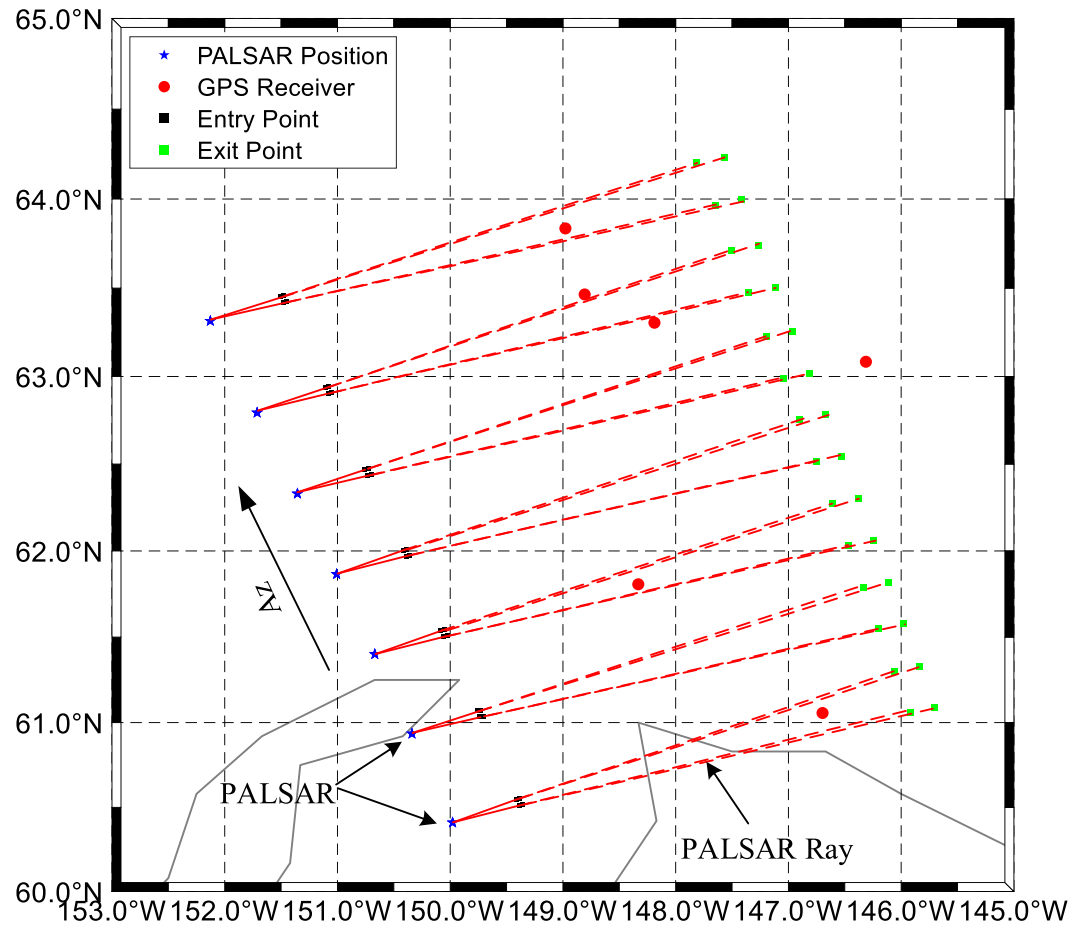


Figure 3. Geographic relationship of the 7 positions of PALSAR (blue pentagram), 6 GPS receivers (solid red point), intersections of 28 PALSAR rays entering (black squares) and exiting (green squares) the ionosphere, and the voxels of the reconstructed ionosphere.

then the system error and the corresponding true FRA value of each sub-scene are added to the data. The FRA measurements for each ray can therefore be retrieved based on the procedure detailed in Section 2.1. The iterative procedure of Equation 19 is repeated using A' and the FRA measurements to reconstruct the $H_0 \cdot Ne_{SAR}$ distribution. The reconstructed Ne_{SAR} values on the 7 longitude surfaces can then be obtained, which are shown in Figures 5b1–5b7 when H_0 is used for decomposition again. The evaluation standards for CIT precision are made by the average deviation (AD), that is,

$$AD = \text{mean}|x_{j,\text{real}} - x_{j,\text{CIT}}| \quad (23)$$

where $x_{j,\text{real}}$ is the true value of the IED, and the $x_{j,\text{CIT}}$ is the IED of CIT based on GPS or GPS-SAR.

Figure 6 shows the AD between the true distribution (Figure 4) and the reconstructions based on GPS and GPS-PALSAR (Figure 5). The AD on the 7 longitude surfaces are shown in Figures 6a1–6a7 based on GPS and Figures 6b1–6b7 based on GPS-PALSAR. Figures 6c1–6c7 are the projections of the PALSAR ray trajectories within the corresponding longitudinal voxels onto the latitude-altitude plane, which can be used to explain the improvement of accuracy after adding PALSAR data. Correspondingly, Table 1 presents the AD values between the true value and the GPS or GPS-PALSAR CIT results on the 7 longitude surfaces. The improvement on the 147.5 longitude surface is evident. The AD for the GPS-based method is $3.626 \times 10^9 \text{ el/m}^3$, while after incorporating PALSAR data, the AD decreases to $2.881 \times 10^9 \text{ el/m}^3$, which represents an increase in accuracy by 20.612%. Comparing with Figure 6a3, the AD in Figure 6b3 has a significant reduction within the latitude of $60^\circ\text{N} \sim 61^\circ\text{N}$ and the altitude range of 300–400 km. Additionally, there is a certain improvement below 300 km

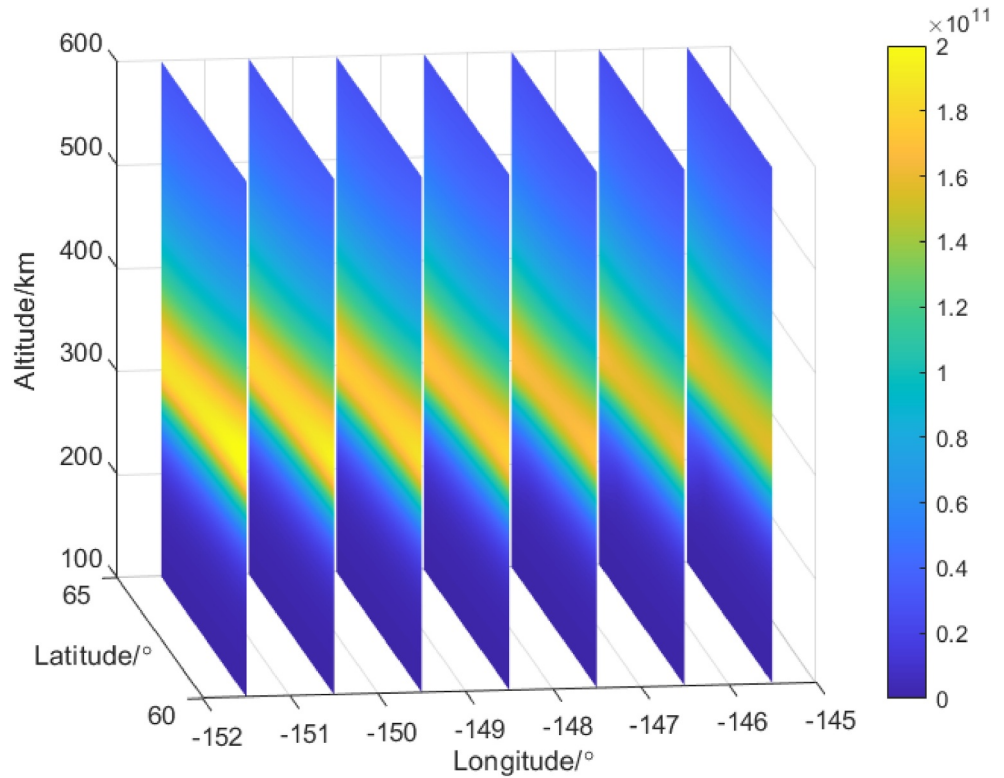


Figure 4. True electron density distribution.

within $61^{\circ}\text{N} \sim 63^{\circ}\text{N}$. This improvement can be explained from Figure 6c3, which shows the projections of the PALSAR ray trajectories within the voxels between 147°W and 148°W onto the latitude-altitude plane. The areas where the corresponding AD decreases correspond to regions where PALSAR rays penetrate, suggesting that the voxel electron density in those regions has undergone correction. While other regions with large deviation are not traversed by any PALSAR rays. On the longitudes of 146.5°W , 148.5°W , and 149.5°W , the accuracy has been improved to a certain extent after adding PALSAR data, which are 5.608%, 4.753%, and 5.017%, respectively, which can also be explained from the corresponding PALSAR ray trajectories (i.e., Figures 6c2, 6c4, and 6c5). In the areas where rays pass through, the AD value decreases. However, since the GPS-based results do not have a large deviation in the areas where the PALSAR rays pass through, the further improvement of the accuracy in these areas after adding PALSAR rays cannot play a decisive role in improving the accuracy of the entire reconstruction area. On the 145.5°W , 150.5°W , and 151.5°W longitude surfaces, there are almost no change in AD between the GPS and GPS-PALSAR. The ray trajectories reveal that there are virtually no rays passing

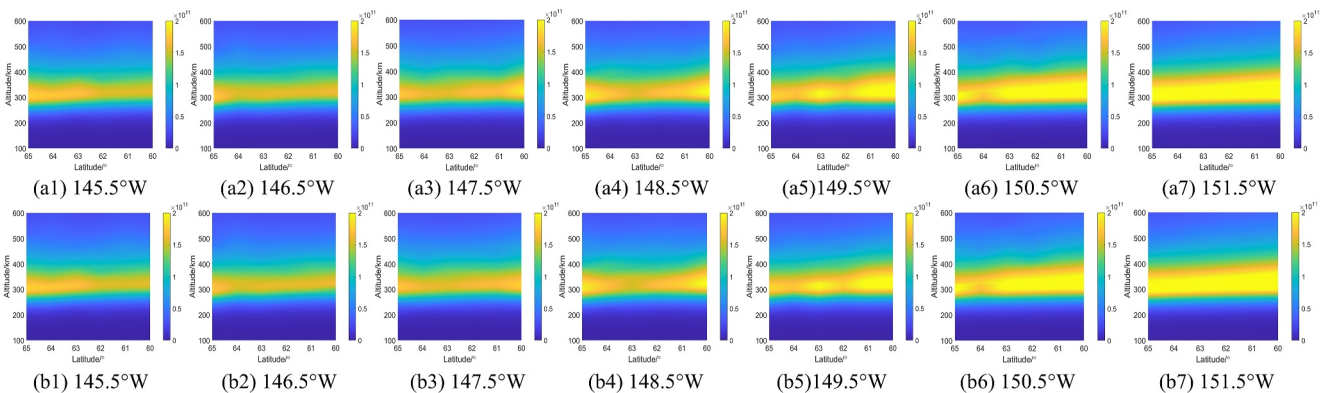


Figure 5. Computerized ionospheric tomography reconstructions based on GPS (a1 to a7) and GPS-PALSAR (b1 to b7) on the different longitude surfaces.

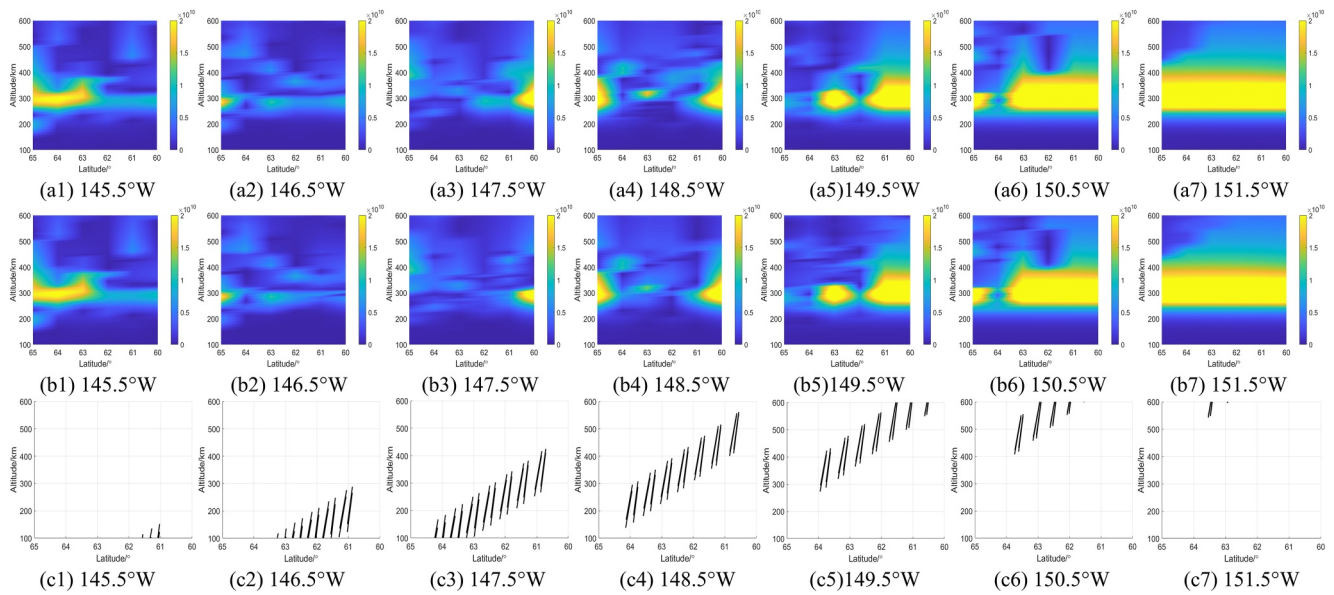


Figure 6. Average deviations between the true distribution (i.e., Figure 4) and reconstructions (i.e., Figure 5) based on GPS (a1 to a7) and GPS-PALSAR (b1 to b7) on the different longitude surfaces. The c1 to c7 are the projections of the PALSAR ray trajectories within the corresponding longitudinal voxels onto the latitude-altitude plane.

through these three surfaces, with only a handful of voxels featuring small AD undergoing correction by the PALSAR rays. However, the impact on the entire reconstruction accuracy can be ignored. The theoretical explanation is that when $a_{ij} = 0$ in Equation 17, indicating that a certain voxel is not traversed by any ray, $\mathbf{X}_j^{(k+1)} = \mathbf{X}_j^{(k)}$. Thus, the reconstructed electron density remains unmodulated and retains its initial value. If these voxel portions are traversed by rays, that is, $a_{ij} \neq 0$, they will be corrected.

Compared the numerous GPS satellites, PALSAR only has one satellite. When PALSAR performs IED reconstruction on a selected area, the beam can only illuminate the area from a specific angle for a short period of time, resulting in limited observations of the scene (i.e., limited number of rays). As demonstrated in the simulation case in Figure 3, a total of 4×7 PALSAR rays are incorporated into tomographic imaging. Unlike GPS, which can receive signals from a variety of angles at various ground receiver stations, PALSAR cannot provide such angle diversity during the imaging process. Although dividing the scene into more sub-scenes can generate more rays, such division will fail to a certain extent. This is because smaller divisions will cause adjacent ray paths to converge and pass through the same voxel, which is the invalid ray for CIT. In order to further improve the accuracy, it is necessary to simultaneously obtain observational data from different off-nadir angles. However, PALSAR does not operate in such a mode. Although the current working mode of PALSAR may not achieve the desired CIT concept, it can still serve as an effective supplement to GPS-based tomographic imaging. The

Table 1

Average Deviation of Computerized Ionospheric Tomography Results in Figure 6

Longitude (°)	AD of GPS ($1 \times 10^9 \text{ el/m}^3$)	AD of GPS-PALSAR ($1 \times 10^9 \text{ el/m}^3$)	Improvement (%)
145.5	3.712	3.711	0.027
146.5	2.425	2.289	5.608
147.5	3.629	2.881	20.612
148.5	4.208	4.188	4.753
149.5	5.362	5.093	5.017
150.5	7.845	7.784	0.778
151.5	10.213	10.206	0.069

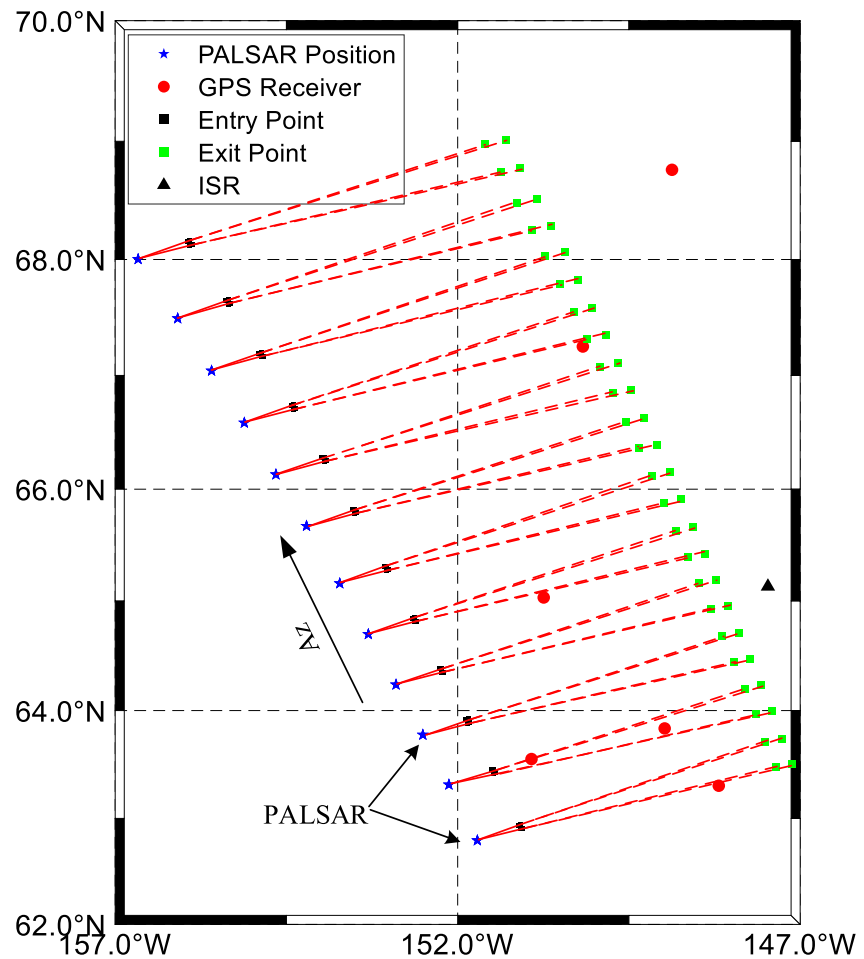


Figure 7. Geographic relationship of the 12 positions of PALSAR (blue pentagram), 6 GPS receivers (solid red point), intersections of 48 PALSAR rays entering (black squares) and exiting (green squares) the ionosphere, the ISR (black triangle) and the voxels of the reconstructed ionosphere.

evidence is in the results shown in Figures 5 and 6, and Table 1, essentially because the 28 PALSAR rays further modulate the IED of the voxels they pass through.

3.2. Measured Data Experiment

As shown in Figure 7, the reconstruction area is set to 147° to 152°W longitude, 62° to 70°N latitude, and 100–600 km altitude, with a voxel size of $5^\circ \times 2^\circ \times 10\text{km}$. Within the reconstruction area, 12 scenes of PALSAR radar data (marked by blue pentagram, data name: ALPSRP276021280 to ALPSRP276021390) along ascending orbit over Alaska on 31 March 2011 are used for ionospheric reconstruction. Similar to Figure 3, the 12 PALSAR coordinate points correspond to 12 radar images and 48 intersections of PALSAR rays entering (black squares) and exiting (green squares) the ionosphere. After removing the receiving stations that are close to each other, there are 6 GPS receiving stations in the reconstruction area (marked by solid red point), and 533 valid rays are received. The GPS ray information and TEC measurement values can be obtained from the GPS file. Based on the results of the above PALSAR and GPS data processing, the IED reconstruction could be performed according to the CIT process (i.e., Figure 2). In this paper, we select the IED value of the incoherent scatter radar (ISR) near the Poker Flat, Alaska (147.47°W, 65.13°N, marked by black triangle) as the true value for comparison. This is because ISR is currently the most powerful and accurate means of detecting the ionosphere. Compared with the other data source, its results are more suitable for comparison as true values. The PALSAR observations are made between about 07:27 and 07:29 UT on 31 March 2011, and all the radar observations are acquired at a nadir angle of 21.5°. The GPS data from 07:26 UT to 07:30 UT on 31 March 2011 in the corresponding area are selected for

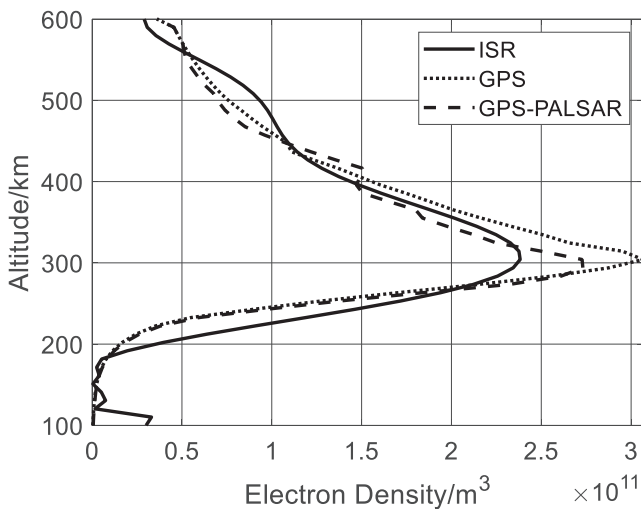


Figure 8. ISR retrieval and ionospheric electron density reconstructions based on GPS and GPS-PALSAR.

CIT. In addition, the ISR data time is 07:28 UT on 31 March 2011. We can see that the setting of the above parameters allows us to obtain IED reconstruction results at almost the same time and location, which is helpful for comparing the two CIT algorithms.

The 1-D reconstructions of IED profile in the altitude direction of the voxels where the ISR is located (i.e., between 147°W and 152°W in longitude and 64°N and 66°N in latitude) are shown in Figure 8. The solid line is the ISR result and regarded as the true value. The thin and thick dashed lines represent the GPS-based and GPS-PALSAR-based tomography results, respectively. The AD value of the GPS-based reconstruction result is $1.88 \times 10^{10} \text{ el/m}^3$, while the AD value of the GPS-PALSAR reconstruction result is $1.71 \times 10^{10} \text{ el/m}^3$, with an accuracy improvement of 9.04%. Although the addition of PALSAR data has improved the accuracy of GPS-based tomography to a certain extent, there are still differences from the true value, that is, the results of ISR. As discussed in Section 2, for GPS-PALSAR-based CIT, the reconstruction accuracy is mainly controlled by factors such as the initial value and the number of PALSAR rays, which are the main reason for the difference between the reconstruction results and ISR. As shown in Figure 7,

the limitation of the number of PALSAR rays (i.e., 48) cannot correct the voxels at all altitudes. In addition, the greater the difference between the initial value (i.e., the GPS-based output) and the true value, the lower the accuracy of GPS-PALSAR-based tomography. However, even so, after incorporating PALSAR data, the reconstruction result is closer to the true value, especially near the peak height, indicating that the corresponding voxels have been corrected by the PALSAR rays. The above conclusions are consistent with the simulation conclusions in Section 3.1.

4. Conclusions

CIT is a technique that uses the integral effect (TEC or FRA) of the ionosphere on electromagnetic waves to reconstruct high-dimensional IED distributions. The data for GPS-based CIT is not comprehensive due to limited perspectives and sparse station distributions. Some voxel electron densities are unaffected by reconstruction because the GPS rays do not pass through them. As a consequence, the tomographic precision is low. Aiming at this issue and combining the characteristics of FRA, this paper proposes a secondary CIT algorithm that combines GPS and full-pol PALSAR data, which can supplement the FRA information retrieved by the PALSAR into CIT. The incorporation of PALSAR data into tomographic imaging introduces new angle ray information, enabling modulation of a greater number of voxel electron densities, thereby enhancing the accuracy of CIT. The simulation results indicate that the tomography results are substantially enhanced in regions where PALSAR rays pass through. The accuracy can be improved up to about 20% in areas with more PALSAR rays passing through. Similar conclusion is obtained from the measured data, with about 9% increase in accuracy.

Although the PALSAR can improve the GPS-based CIT accuracy to a certain extent, the current PALSAR working mode still restricts the ray information supplemented to tomography. For instance, only a limited number of PALSAR rays can be obtained within the imaging swath width under a fixed off-nadir angle condition, and the voxel electron density cannot be increased on a large scale. Therefore, future work will try to use more L-band SAR satellite data, such as LuTan-1 (Liu et al., 2022), which contains two full-pol L-band SAR, to improve the limitations of SAR in CIT.

Data Availability Statement

The GPS and PALSAR data can be available at Zhao (2024a, 2024b).

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